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AIM: Implement logistic Regression learning algorithm.

Dataset: Breast Cancer

```
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load dataset
from sklearn.datasets import load_breast_cancer

# Split dataset
from sklearn.model_selection import train_test_split

# Feature scaling
from sklearn.preprocessing import StandardScaler

# Logistic Regression model
from sklearn.linear_model import LogisticRegression

# Evaluation metrics
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import seaborn as sns
```

```
# Load breast cancer dataset
data = load_breast_cancer()

# Features (input variables)
X = data.data

# Target labels (0 = malignant, 1 = benign)
y = data.target

# Convert dataset into DataFrame
df = pd.DataFrame(X, columns=data.feature_names)

# Add target column
df["target"] = y

# Display first few rows
print(df.head())

# Total records
print("Total records:", df.shape[0])
```

```
   mean radius  mean texture  mean perimeter  mean area  mean smoothness  \
0        17.99         10.38          122.80     1001.0          0.11840
1        20.57         17.77          132.90     1326.0          0.08474
2        19.69         21.25          130.00     1203.0          0.10960
3        11.42         20.38           77.58      386.1          0.14250
4        20.29         14.34          135.10     1297.0          0.10030

   mean compactness  mean concavity  mean concave points  mean symmetry  \
0           0.27760          0.3001              0.14710         0.2419
1           0.07864          0.0869              0.07017         0.1812
2           0.15990          0.1974              0.12790         0.2069
3           0.28390          0.2414              0.10520         0.2597
4           0.13280          0.1980              0.10430         0.1809

   mean fractal dimension  ...  worst texture  worst perimeter  worst area  \
0                 0.07871  ...          17.33           184.60      2019.0
1                 0.05667  ...          23.41           158.80      1956.0
```

```
2               0.05999  ...          25.53           152.50      1709.0
3               0.09744  ...          26.50            98.87       567.7
4               0.05883  ...          16.67           152.20      1575.0

   worst smoothness  worst compactness  worst concavity  worst concave points  \
0            0.1622             0.6656           0.7119                0.2654
1            0.1238             0.1866           0.2416                0.1860
2            0.1444             0.4245           0.4504                0.2430
3            0.2098             0.8663           0.6869                0.2575
4            0.1374             0.2050           0.4000                0.1625

   worst symmetry  worst fractal dimension  target
0          0.4601                  0.11890       0
1          0.2750                  0.08902       0
2          0.3613                  0.08758       0
3          0.6638                  0.17300       0
4          0.2364                  0.07678       0

[5 rows x 31 columns]
Total records: 569
```

```
# Split data into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

```
# Standardize features

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
# Create logistic regression model

model = LogisticRegression(max_iter=1000)

# Train the model
model.fit(X_train, y_train)
```

```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
# Predict using test data

y_pred = model.predict(X_test)
```

```
# Accuracy of model
print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred))

# Detailed classification report
print(classification_report(y_test, y_pred))
```

```
Logistic Regression Accuracy: 0.9736842105263158
              precision    recall  f1-score   support

           0       0.98      0.95      0.96        43
           1       0.97      0.99      0.98        71

    accuracy                           0.97       114
   macro avg       0.97      0.97      0.97       114
weighted avg       0.97      0.97      0.97       114
```

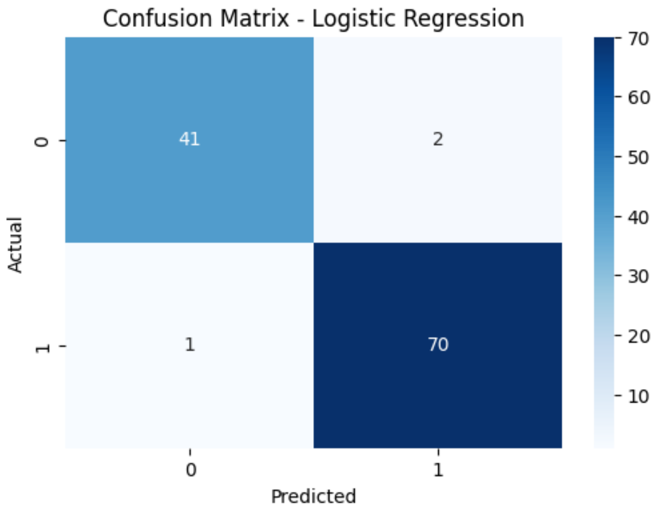
```
# Generate confusion matrix
cm = confusion_matrix(y_test, y_pred)
```

```
# Plot confusion matrix
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, cmap="Blues")

plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.title("Confusion Matrix - Logistic Regression")

plt.show()
```




```
# Example new data prediction

sample = [X_test[0]]

prediction = model.predict(sample)

if prediction[0] == 0:
    print("Prediction: Malignant Tumor")
else:
    print("Prediction: Benign Tumor")
```

```
Prediction: Benign Tumor
```

Inference: In this experiment, the Logistic Regression algorithm was implemented using the Breast Cancer dataset. The dataset was split into training and testing sets, and feature scaling was applied to improve model performance. Logistic regression predicts the probability of class membership using the sigmoid function. The model was evaluated using accuracy, classification metrics, and a confusion matrix. The results show that logistic regression effectively classifies tumors as malignant or benign with high accuracy.